ASHVINY ARUMUGAM

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PROFILE:

I am a professional Electronics Engineer with over 2 years of experience in Advantest Verigy 93K tester platform using Advantest Smartest 8 in semiconductor industry. I had the opportunity to drive several projects which helped me to enhance both technical and managerial skills. I am looking forward to continuing my career in an organization that utilizes my knowledge and experience.

TECHNICAL SKILLS:

Category	Tools and Technologies
ATE & Validation tools	Advantest 93K, Smartest 8, Exensio, O Plus
Debug & Analysis Tools	Python, Power BI, MS Excel
Database & Data Handling	MySQL (queries, data modeling)
Programming/Scripting Languages	Python, C
Digital Design & Simulation	Verilog HDL
Version Control	SVN, Git
OS Environments	Microsoft Windows, Linux (Red Hat CentOS)

ONLINE TRAINING:

Maven Silicon / Maven Analytics – Design for Test (DFT) Training **2026 – Present**

- Developed theoretical knowledge in Scan, ATPG, fault models, MBIST, JTAG/IJTAG, test compression, and DFT DRC
- Gained understanding of scan architecture, stuck-at/transition faults, fault coverage, BIRA/BISR, and boundary-scan concepts.

WORK EXPERIENCE:

Validation Engineer (Silicon), Tessolve Ltd (Qualcomm consultant) **2021 – 2023**

- Developed and debugged test programs on **Advantest V93K** platforms using **Smartest 8**, supporting silicon bring-up for production and engineering samples.
- Conducted **post-silicon validation** across multiple stages including **wafer sort** and **final test**, enabling robust screening and device qualification.
- Performed debug and characterization of **structural blocks (ATPG, TDF, MBIST)** across **process, voltage, and temperature (PVT)** corners to ensure product reliability.
- Used **shmoo plotting**, **test tables**, and **data analytics tools (Exensio, O Plus)** to identify bottlenecks, improve first-pass yield, and reduce test time.
- Leveraged **version control tools** (SVN, Git) to manage program revisions, ensuring traceability and efficient rollout across production builds.
- Collaborated cross-functionally with **design, DFT, and product engineering teams** to drive **root cause analysis (RCA)** and implement test escapes prevention strategies.
- Supported customer return analysis (**RMA**) by performing failure analysis and regression validation to expedite closure cycles.
- Contributed to yield monitoring, **statistical data analysis**, and test coverage optimization.

EDUCATION:

- **Nanyang Technological University**, Singapore **Aug 2020 – Aug 2021**
  Master of Science, Electronics Engineering
- **Anna University - MIT**, Chennai, India **July 2015 – May 2019**
  Bachelor of Engineering, Electronics and Instrumentation

PROJECTS:

Cryptocurrency Analysis (Capstone Project)
TeqCertify Data Analysis Training **Nov 2024 – April 2025**

- Used Power BI tool to analyse, visualize data, creatin interactive dashboards for insights to track trends, patterns and make data driven decisions efficiently.

Design & implementation of JFET based colour sensors
Nanyang Technological University, Singapore **Oct 2020 – Jun 2021**

- Examination and realization of optical and colour sensing in silicon.
- Analysing the behavioural changes in JFET when used as photosensor.
- Calculation of certain optical characteristics namely optical conductivity, absorption coefficient, full width half maximum, attenuation loss and resistivity in order to analyse its sensing performance.
- In addition, theoretical calculations for the fabrication of proposed structure were made.

Three axis control of a pen using PLC.
Anna University-MIT, India **Aug 2018 – April 2019**

- Design a three-axis control system to control the robotic hand containing a pen which could draw on a sheet of paper.
- The entire system is controlled by MITSUBISHI-PLC and the software used was GX Developer.

Detector Wireless controlled bot using Arduino
Anna University-MIT, India **Aug 2017 – Feb 2018**

- Worked in building DTMF (Dual tone multi frequency) controlled bot to capture live footage of the environment using a mobile phone or laptops which would travel in places where human intervention is risky.
- DTMF controlled module: MT8870DE
- Microcontroller platform: Arduino.

WORKSHOPS AND TRAININGS

- Control system training in Salem Steel Plant, India, 2016.
- Security system training in Chennai International airport, India, 2017.
- PIC-Embedded Workshop on MSP430, Anna University, India. 2018